

14. **Ans: D**

$$\begin{aligned}V_{\max} &= \omega X_0 \\&= 2\pi f X_0 \\&= 2\pi \frac{v}{\lambda} X_0 \\&= 2\pi \times \frac{4.0}{0.15} \times 0.030 \\&= 5.0 \text{ m s}^{-1}\end{aligned}$$

15. **Ans: B**